

Correspondences and fixed points

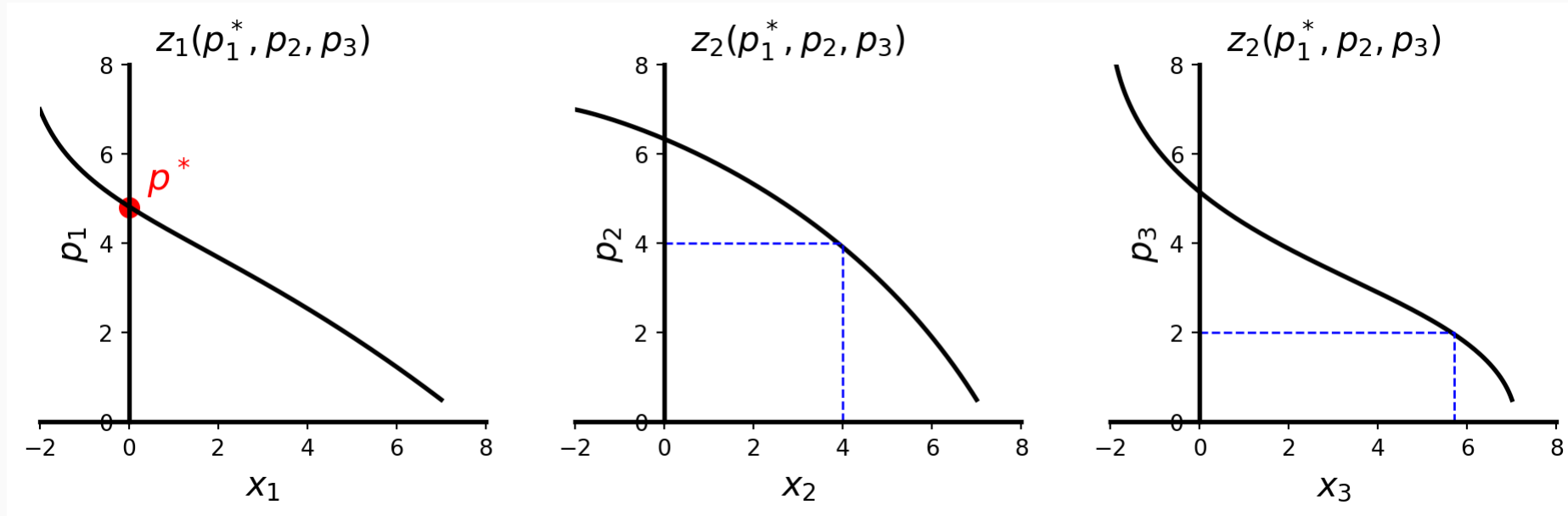
A pure exchange economy

- A pure exchange economy with three goods.
- Excess demand good $\ell = 1, 2, 3$ is

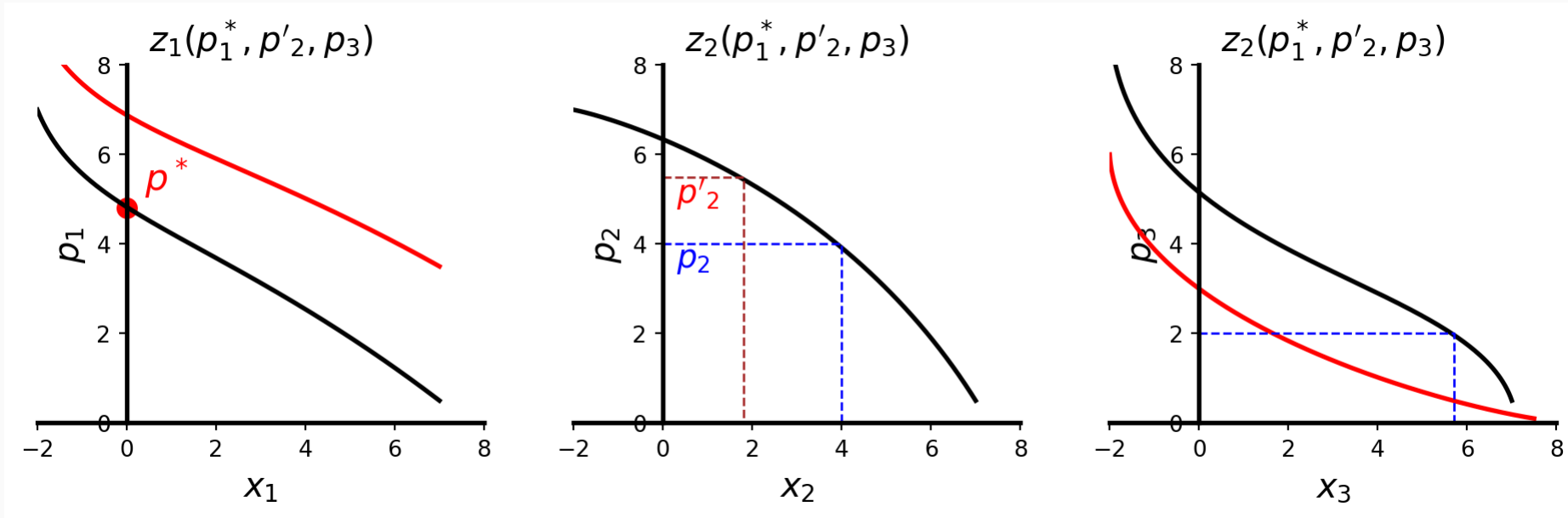
$$z_{\ell}(p_1, p_2, p_3) = D_{\ell}(p_1, p_2, p_3, \omega_{\ell}) - \omega_{\ell}$$

where D_{ℓ} is the aggregate demand for good ℓ , the sum of the individual demands of agents in the economy.

The difficulty



Increase p_2 to p'_2



Fixed points and correspondences review

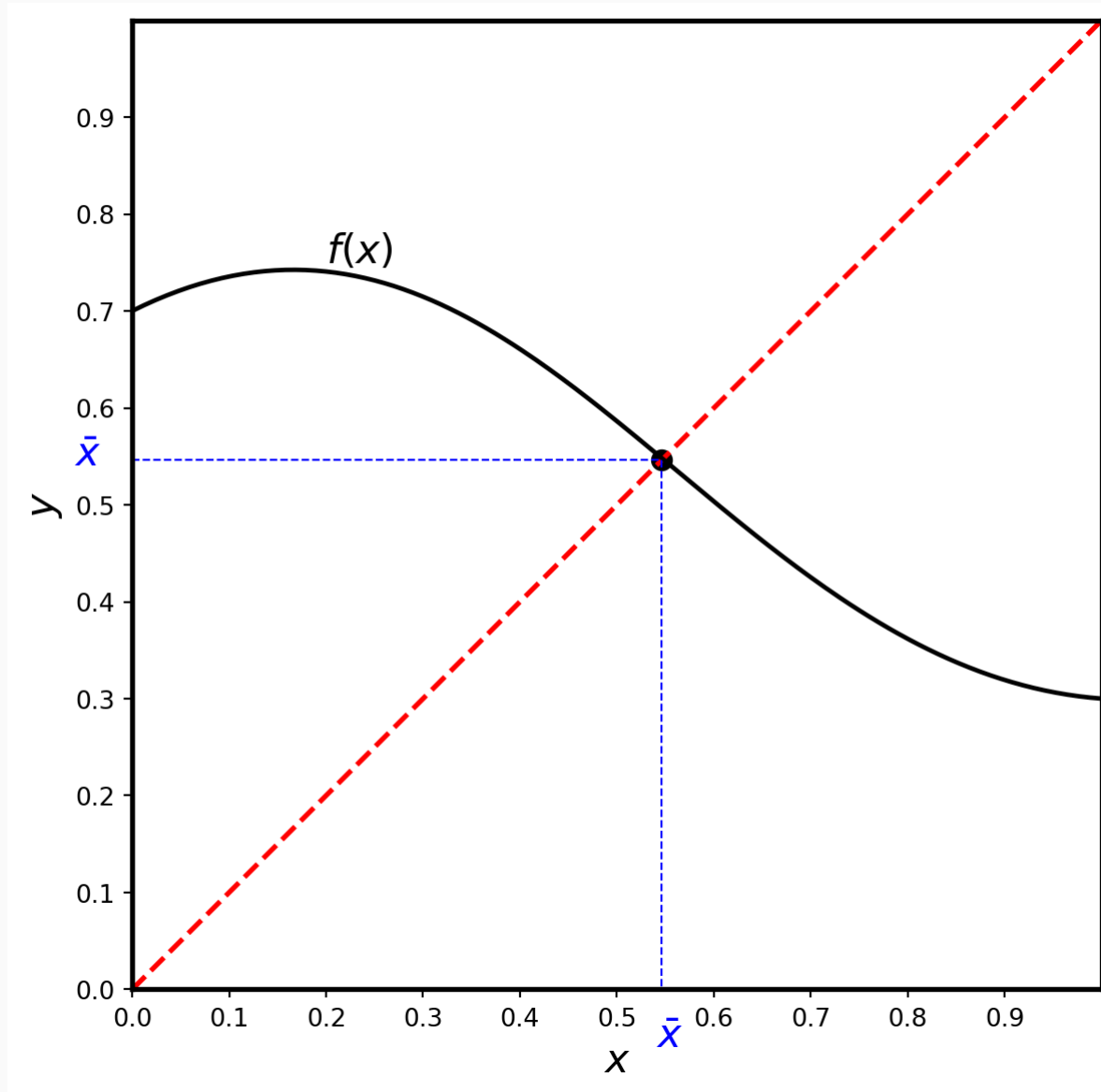
Fixed point of a function

Definition 1 (Fixed point) Let $f : X \rightarrow X$. $x \in X$ is a fixed point of f if $f(x) = x$.

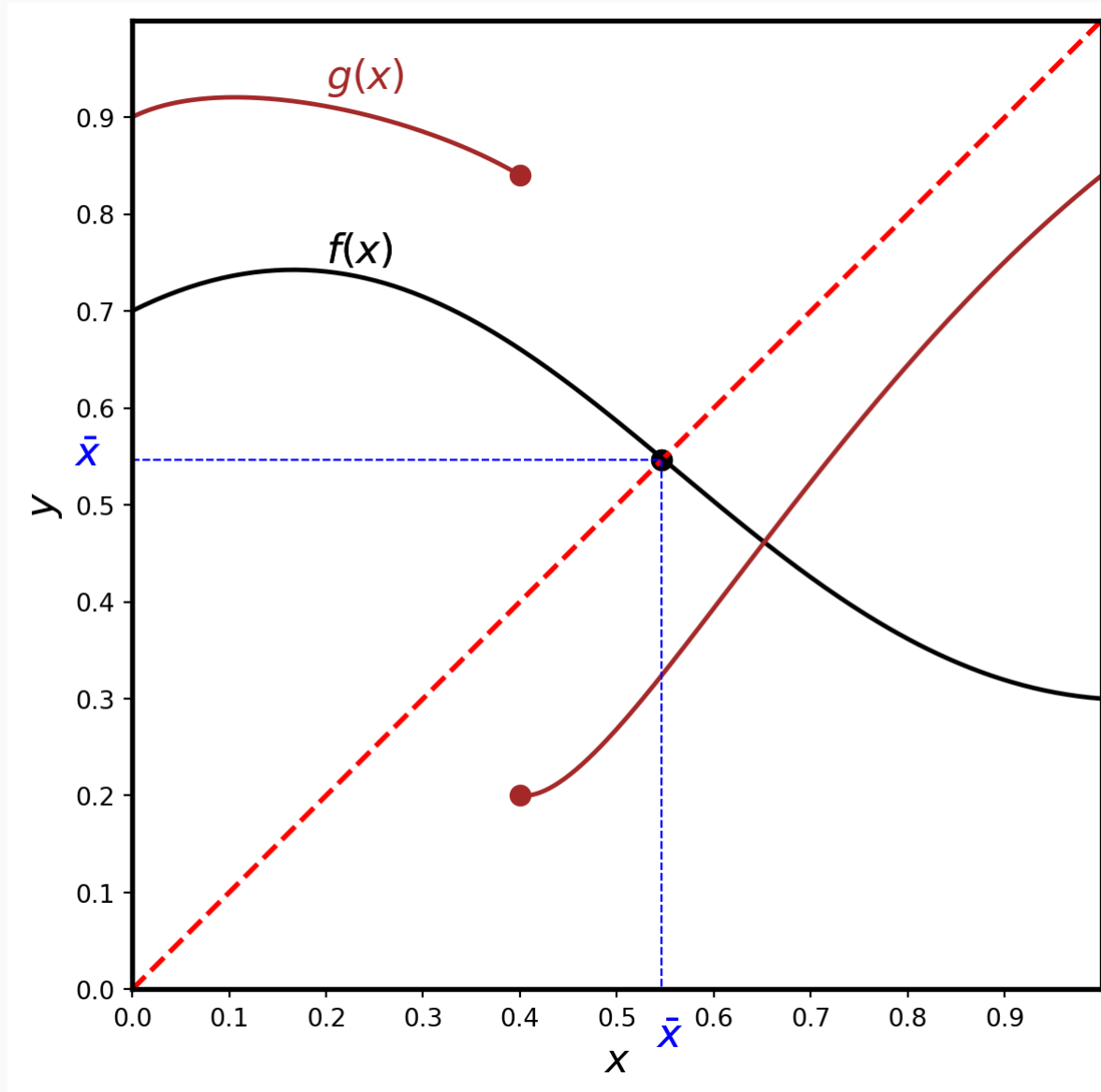
Theorem 1 (Brouwer) Let $K \subseteq \mathbb{R}^n$ be a compact, convex set and $f : K \rightarrow K$ be a continuous function. Then $(\exists x \in K) f(x) = x$.

- If $X \subset \mathbb{R}$, **Theorem 1** is a consequence of the intermediate value theorem.

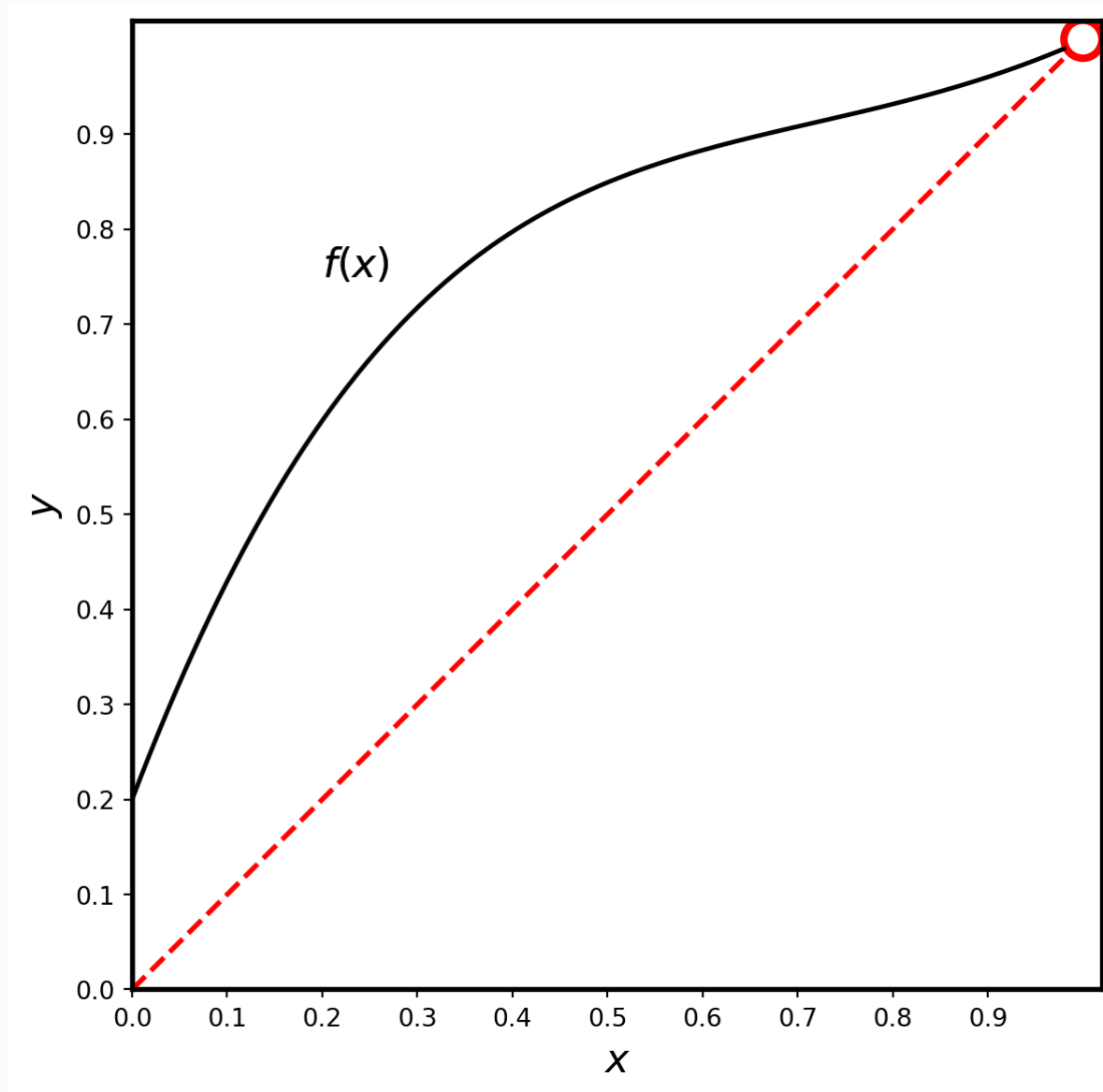
Example Brouwer FPT



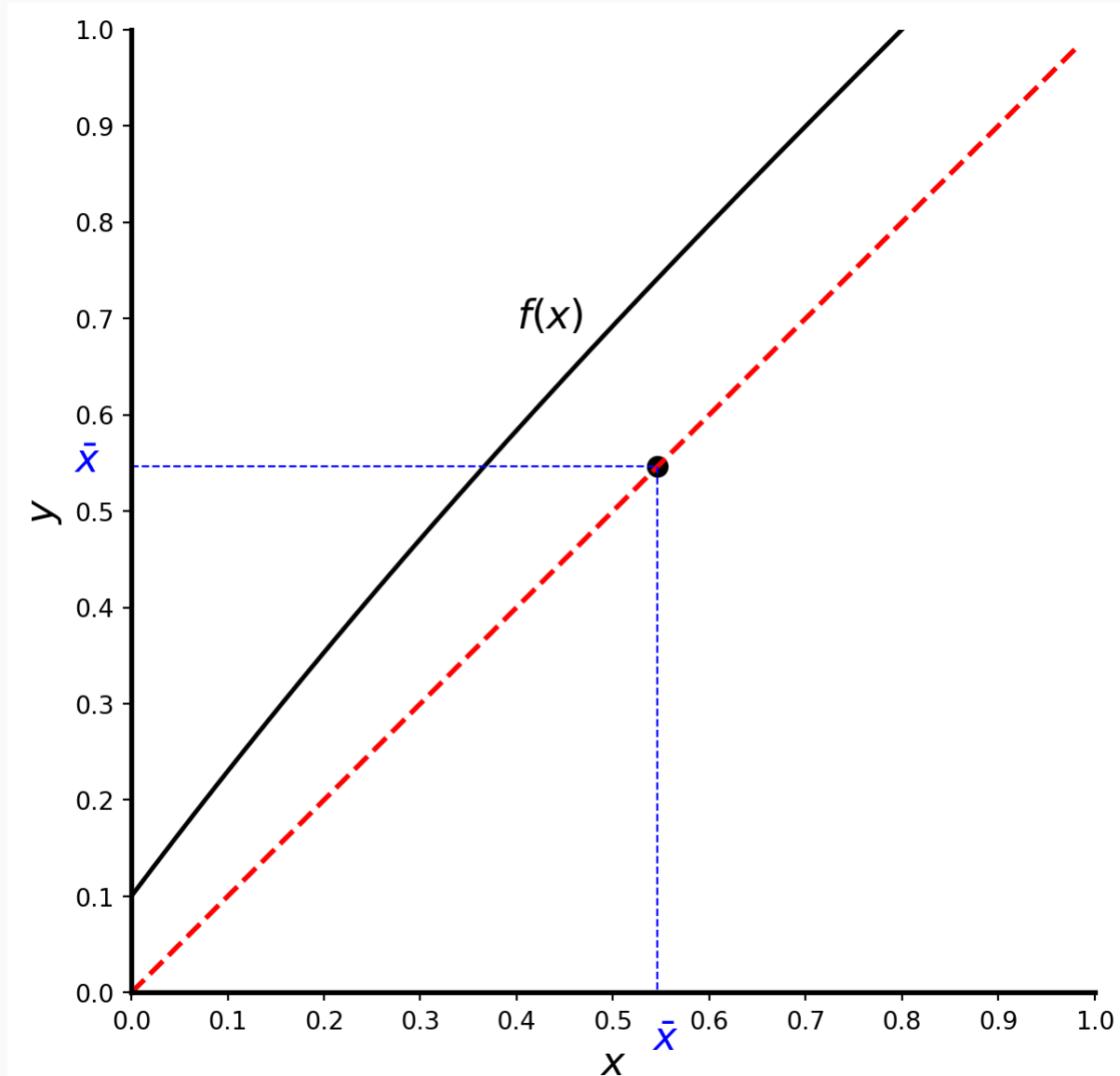
Example discontinuous function



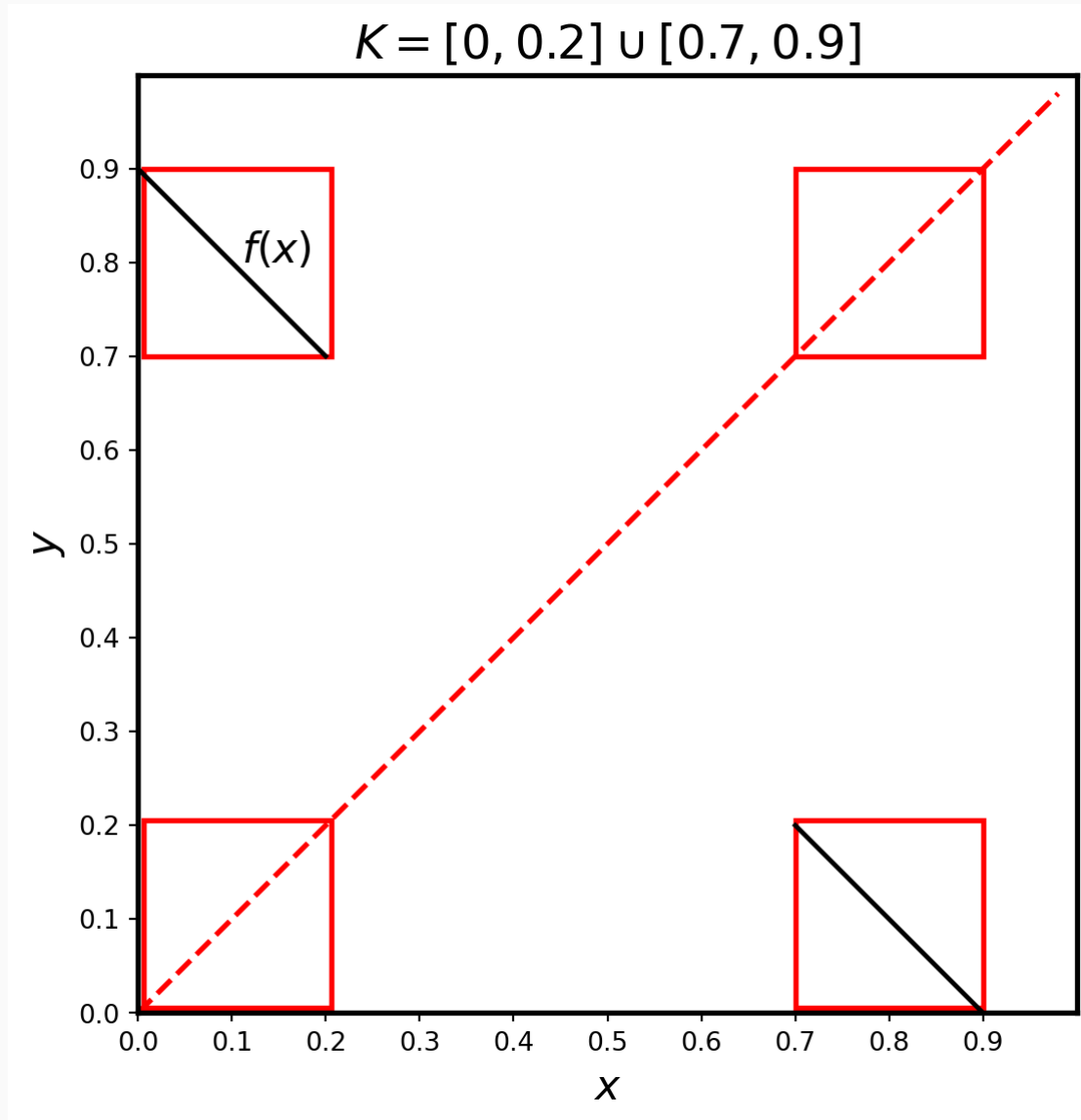
Example set K is not closed



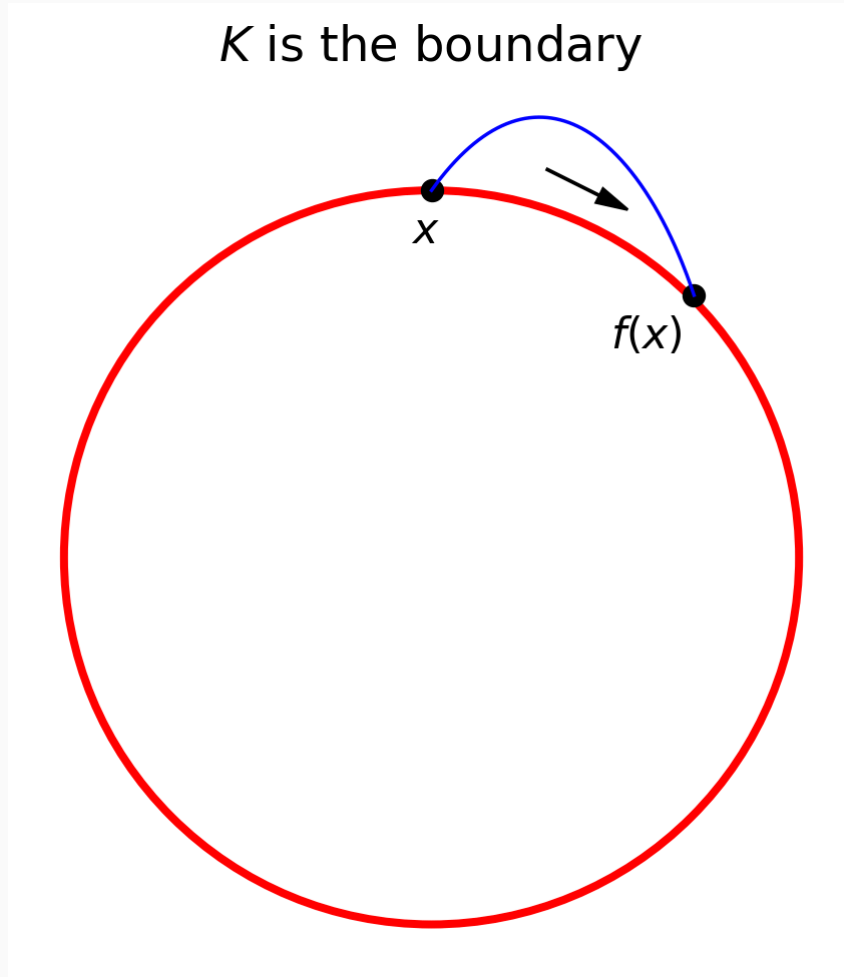
Example set K is not bounded



Example set K is not convex



Example K is the boundary of a circle



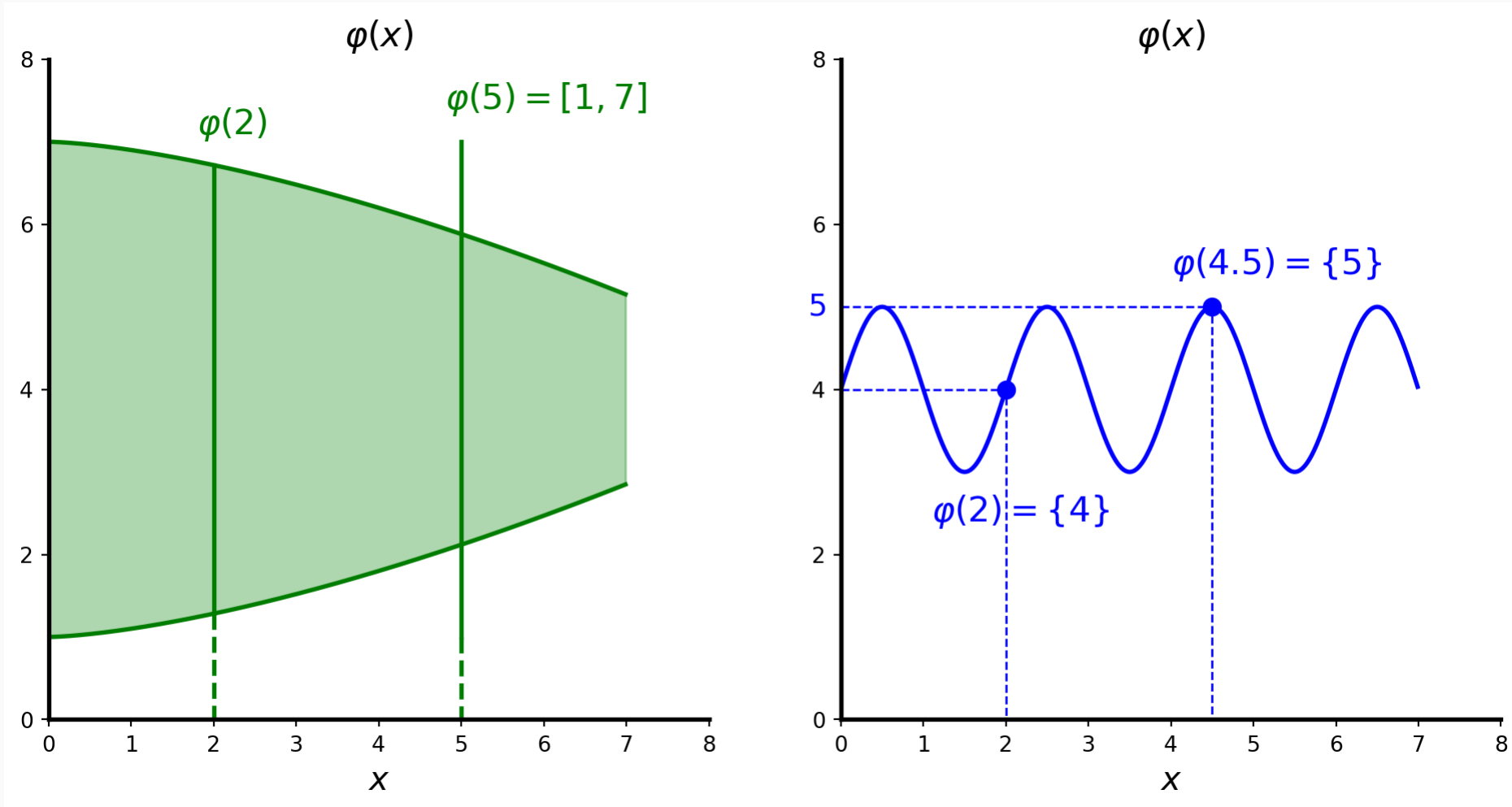
- $f(x) = x + \delta$ where δ represents a fixed arc distance.

Correspondence

Definition 2 (Correspondence) Let X, Y be two sets. A correspondence φ from X to Y , denoted $\varphi : X \rightarrow Y$, is a function that maps each element $x \in X$ to a subset $\varphi(x)$ of Y .

- φ can be seen as a function $\varphi : X \rightarrow 2^Y$, φ maps $x \in X$ to $\varphi(x) \subseteq Y$.
- **Notation.** If X, Y are metric spaces (or topological spaces), τ_X and τ_Y are the collections of open subsets of X and Y respectively.
 - $U \in \tau_X$ means U is an open subset of X .

Example: correspondence



Kakutani FPT

Fixed point of a correspondence

Definition 3 (Fixed point) Let $\varphi : X \rightarrow X$ be a correspondence. $x \in X$ is a fixed point of φ if $x \in \varphi(x)$.

Theorem 2 (Kakutani) Let $K \subseteq \mathbb{R}^n$ and $\varphi : K \rightarrow K$ be a correspondence. If K is compact and convex, and $(\forall x \in K)$

1. $\varphi(x) \neq \emptyset$,
 2. $\varphi(x)$ is convex,
 3. $\varphi(x)$ is closed,
 4. φ is upper-hemicontinuous,
- then

$$(\exists x \in K) \ x \in \varphi(x)$$

Upper-hemicontinuity (uhc)

Definition 4 Let X, Y be metric spaces and $\varphi : X \rightarrow Y$ a correspondence. φ is upper hemi-continuous at $\bar{x}$ if

- $\varphi(\bar{x}) \neq \emptyset$ and
- $(\forall V \in \tau_Y, V \supseteq \varphi(\bar{x})) (\exists U \in \tau_X, U \ni \bar{x})$ such that

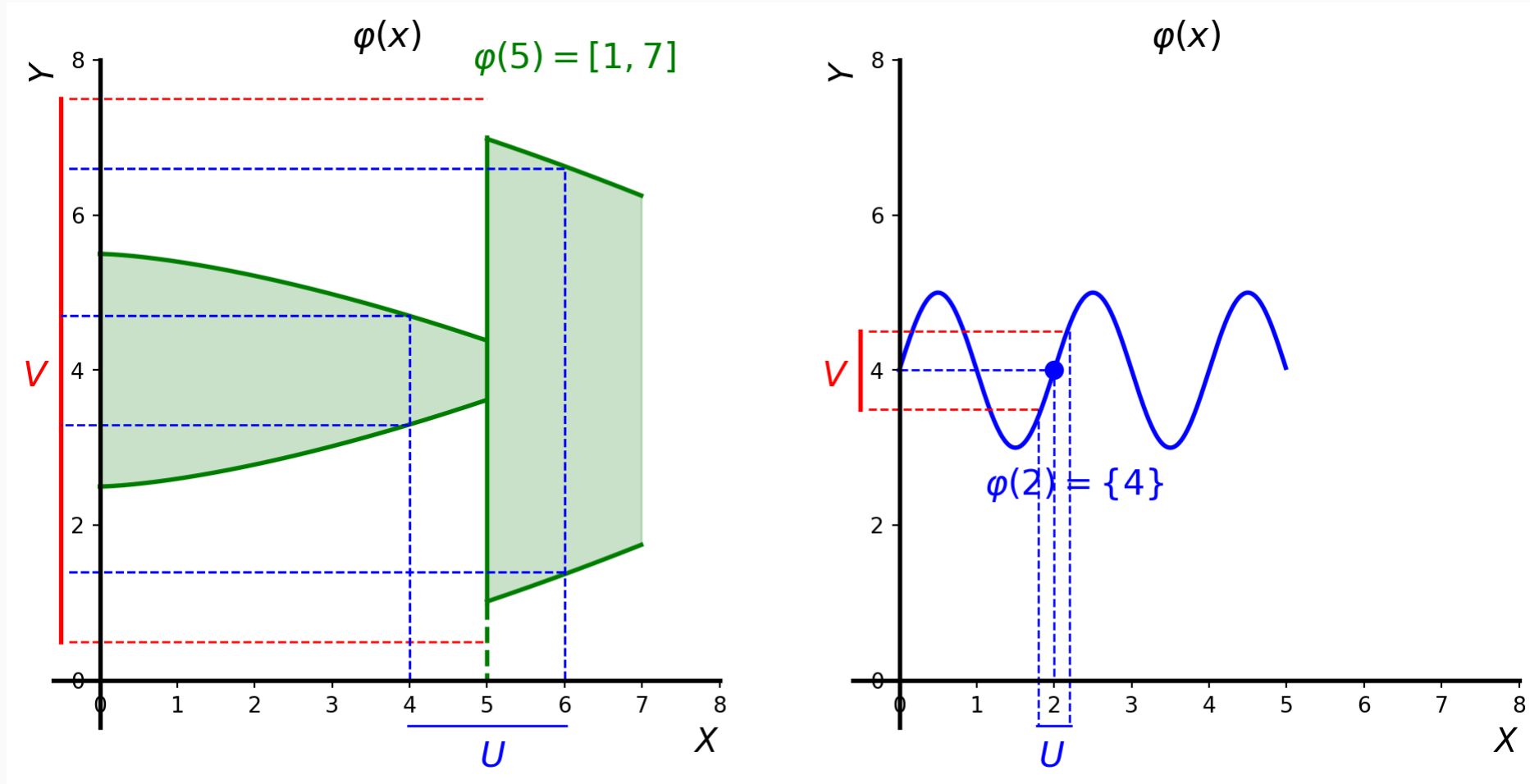
$$[(\forall x \in U) \varphi(x) \subseteq V].$$

φ is upper-hemicontinuous (uhc) if $(\forall x \in X) \varphi(x)$ is uhc at x .

uhc and continuity of functions

Lemma 1 Let X, Y be metric spaces and $\varphi : X \rightarrow Y$ a correspondence. Suppose $(\forall x \in X) \varphi(x) = \{y_x\}, y_x \in Y$. Define $f(x) := y_x$. Then, $\varphi(x)$ is uhc if and only if f is continuous.

Example: uhc correspondence



Lower-hemicontinuous (lhc)

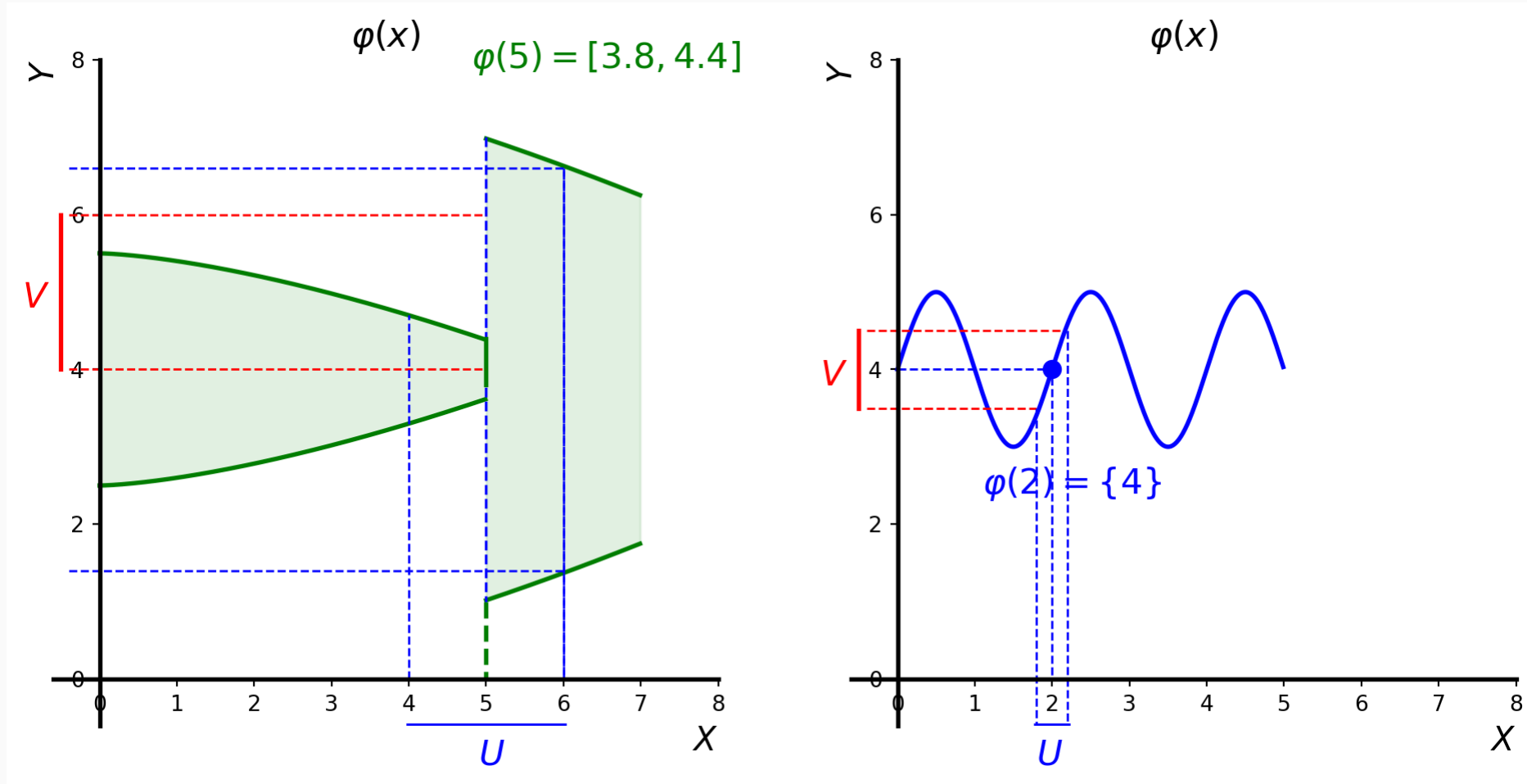
Definition 5 (lhc) Let X, Y be metric spaces and $\varphi : X \rightarrow Y$ a correspondence. φ is lower hemi-continuous (lhc) at $\bar{x}$ if

- $\varphi(\bar{x}) \neq \emptyset$ and
- $(\forall V \in \tau_Y, V \cap \varphi(\bar{x}) \neq \emptyset) (\exists U \in \tau_X)$ such that

$$[(\forall x \in U) \varphi(x) \cap V \neq \emptyset].$$

φ is lhc if $(\forall x \in X) \varphi(x)$ is lhc at x .

Example: Ihc correspondence



Closed correspondence

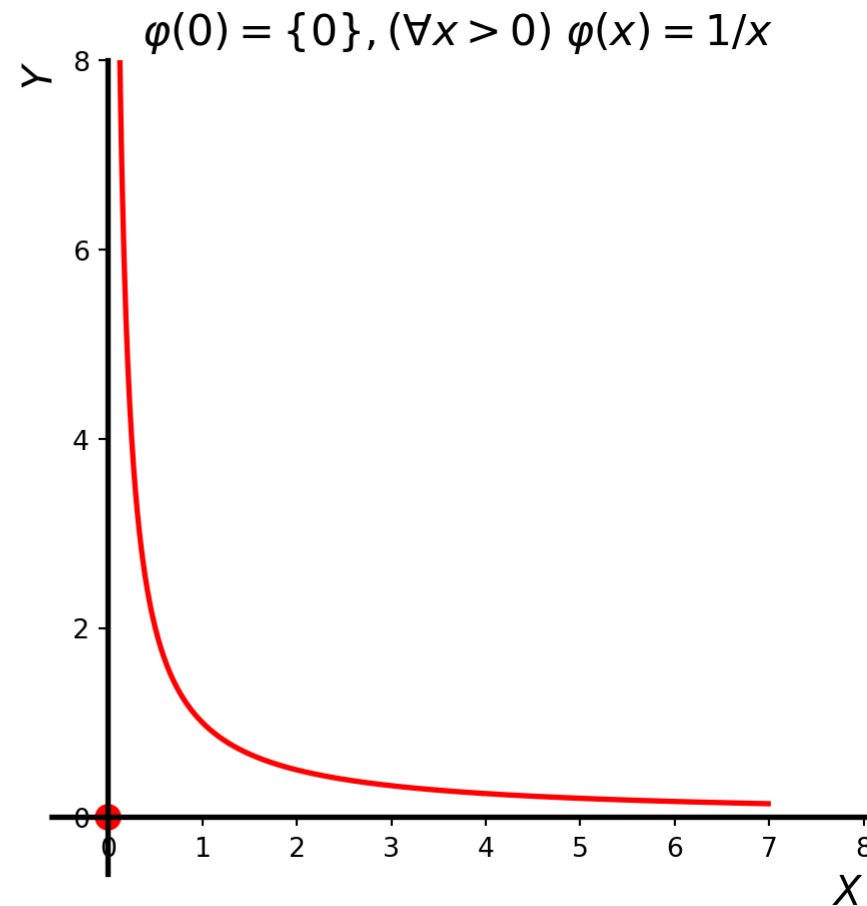
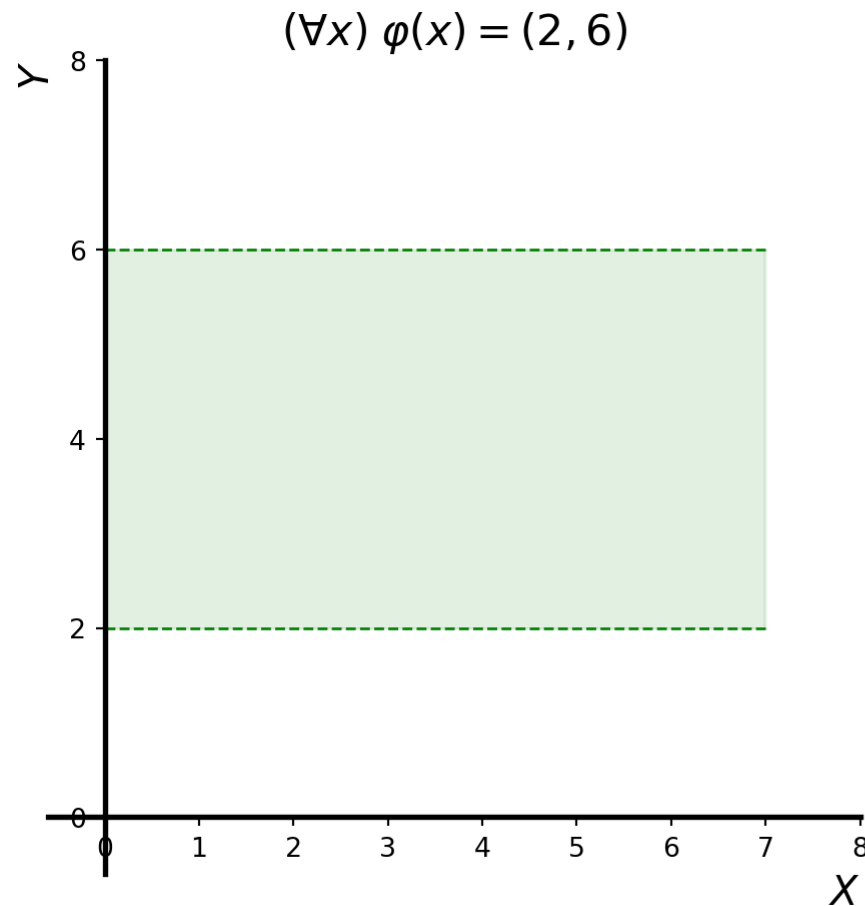
Let X, Y be metric spaces and $\varphi : X \rightarrow Y$ a correspondence. φ is closed if $\text{gr}(\varphi)$ is closed in $X \times Y$ where

$$\text{gr}(\varphi) = \{(x, y) \in X \times Y : y \in \varphi(x), x \in X\}$$

Relation between closed graph and uhc

Theorem 3 Let X, Y be metric spaces and $\varphi : X \rightarrow Y$ a correspondence. If φ is uhc and closed valued then $\text{gr}(\varphi)$ is closed.

2 examples



- Left: φ is uhc but it does not have a closed graph. Right: φ has a closed graph and is closed valued; but not uhc at $x = 0$.

Example

Let $X = \mathbb{R}$, $Y = \mathbb{R}^2$, and $\varphi : X \rightarrow Y$

$$\varphi(x) = \left\{ (y_1, y_2) \in Y : y_2 \geq 0, (\forall y_1 \neq x) \left[y_2 \geq \frac{1}{y_1 - x} \right] \right\}$$

φ is closed-valued and closed, it is not uhc at any $x \in X$.

Theorem 4 Let X be metric spaces, Y a compact metric space, and $\varphi : X \rightarrow Y$ a correspondence. If $\text{gr}(\varphi)$ is closed then φ is uhc and closed-valued.